

		D: Upthrust is a result of the pressure difference at the top and bottom of the cube.
7	B	Assume that the force on top hinge F is pointing horizontally to the right. Taking moments about the bottom hinge, $1.8(200 \cos 30^\circ) + 3(200 \sin 30^\circ) = 1.8 F + 1.5(40)(9.81)$ $F = 12.87 = 13 \text{ N (to the right)}$
8	B	Total output power = power in doing work against friction + power in gaining gravitational